

SIMATS ENGINEERING

CO3- ASSESSMENT TOOL 3

ERROR ANALYSIS TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING

COURSE CODE: ECA09

CO3 – Error Analysis Task

Question 30

Problem Statement

A signal with amplitude range ± 0.9 is scaled by 4 before fixed-point processing. The output saturates. Calculate the expected range after scaling and redesign the scaling factor.

1. INTRODUCTION

Signal scaling is an important operation in Digital Signal Processing (DSP), especially in fixed-point implementations. It adjusts the amplitude of a signal to ensure that it remains within the representable range of the processor. If the scaling factor is too large, the signal exceeds the allowable range, causing overflow and saturation. Saturation results in signal clipping, which introduces distortion and reduces the accuracy of signal processing. Therefore, selecting an appropriate scaling factor is essential for reliable fixed-point DSP systems.

2. GIVEN DATA

- Input Signal Range = ± 0.9
- Scaling Factor = 4
- Fixed-Point Processing Range = ± 1

3. Steps to Solve

Step 1: Calculate the Scaled Signal Range

Formula:

Scaled Signal = Input Signal $\times$ Scaling Factor

Maximum Value:

$$0.9 \times 4 = 3.6$$

Minimum Value:

$$-0.9 \times 4 = -3.6$$

Therefore,

Scaled Signal Range = ± 3.6

Step 2: Identify the Error

The fixed-point processor can represent values only between **-1 and $+1$** .

Since the scaled signal range is **± 3.6** , which is greater than **± 1** , the processor cannot represent the complete signal.

Error Identified:

- Overflow occurs.
- The output becomes saturated (clipped).
- Signal distortion is produced.

Step 3: Redesign the Scaling Factor

To prevent saturation, the following condition must be satisfied:

$$0.9 \times K \leq 1$$

where **K** is the new scaling factor.

Calculation:

$$K \leq 1 / 0.9$$

$$K \leq 1.11$$

Therefore, the **maximum safe scaling factor is 1.11**.

For practical DSP implementation, the scaling factor is chosen as:

$$K = 1$$

Step 4: Verify the Corrected Solution

Using the redesigned scaling factor,

$$0.9 \times 1 = 0.9$$

New Signal Range = ± 0.9

Since ± 0.9 is less than ± 1 , the signal remains within the fixed-point range.

RESULT:

- No Overflow
- No Saturation
- No Signal Clipping

4. FINAL RESULT

- **Original Signal Range:** ± 0.9
- **Scaling Factor Used:** 4
- **Scaled Signal Range:** ± 3.6
- **Error Identified:** Overflow and Saturation
- **Maximum Safe Scaling Factor:** 1.11
- **Recommended Scaling Factor:** 1
- **Corrected Signal Range:** ± 0.9
- **Final Output:** No Overflow and No Saturation

5. CONCLUSION

Scaling the input signal by a factor of **4** increases its amplitude from ± 0.9 to ± 3.6 , which exceeds the allowable fixed-point range of ± 1 . As a result, overflow and saturation occur, causing signal distortion. By redesigning the scaling factor to **1** (or a maximum of **1.11**), the signal remains within the allowable range. This prevents overflow, eliminates saturation, and ensures accurate and reliable fixed-point DSP processing.

FINAL ANSWER

- Expected Signal Range after Scaling = ± 3.6
- Reason for Saturation = The scaled signal exceeds the fixed-point range (± 1), causing overflow and clipping.
- Maximum Safe Scaling Factor = 1.11
- Recommended Practical Scaling Factor = 1
- Corrected Signal Range = ± 0.9
- Final Result = No Overflow, No Saturation, and Improved Signal Quality.